Demographic Bias and Fairness Mitigation in LLM-based Resume Screening: A Literature Review

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Abstract

As Large Language Models (LLMs) become common in recruitment, evidence is mounting that they perpetuate and amplify demographic biases. This report reviews 11 papers that audit LLMs for gender, racial, national origin, disability, and intersectional biases in hiring. The AI screening mechanisms studied here fall into three categories: embedding retrieval, prompt-based scoring, and pairwise ranking. Each exhibits structural biases even after overt demographic markers are removed. Current fairness mitigation strategies are organized into a four layer defense pipeline covering data curation, pre-processing, in-processing, and post-processing. By comparing the methodologies, mathematical guarantees, and empirical limitations of these techniques, this review proposes concrete acceptance criteria for equitable resume screeners and identifies gaps in existing mitigation approaches.

1 Introduction

Recruitment has shifted from manual human review to automated algorithmic screening. Most large corporations now use some form of algorithmic hiring technology, and the market for these tools continues to grow. LLMs such as GPT-4, Claude, Gemini, and Llama now handle resume parsing, candidate ranking, and suitability scoring. Proponents argue that algorithmic hiring is faster and less subjective than human review. But empirical research tells a different story: these systems, trained on historical data that encodes societal prejudices, risk automating discrimination at scale. As the course on Trustworthy Machine Learning emphasizes, accuracy on the training distribution alone is not enough. ML systems must also be *fair*, *robust*, *privacy-preserving*, and *interpretable*. This review shows how all four requirements collide in algorithmic hiring.

The core issue is that vendors often claim their systems are "debiased," but independent audits find otherwise. Biased models produce different suitability scores for resumes with identical qualifications when the only change is a name associated with a different gender, race, national origin, or disability status. Removing names does not fix the problem either, because LLMs pick up on proxy features like career gaps, vocabulary choices, and geographic locations that correlate with protected attributes.

This report surveys 11 research papers to address three questions:

1. How do modern AI screening systems operate, and how do researchers audit them for bias?
2. What are the mathematical mechanisms by which bias persists despite anonymization?
3. What mitigation strategies have been proposed, and what are their respective strengths and limitations?

2 AI Resume Screening Mechanisms

Bias enters resume screening through three main mechanisms.

Embedding Retrieval. Platforms like Amazon convert documents into high dimensional vectors using deep neural networks. The resume embedding $R(d)$ is matched against a job description vector J using cosine similarity:

$$\text{sim}(R, J) = \frac{R \cdot J}{\|R\| \, \|J\|} \tag{1}$$

Bias enters because the vector space inherits historical prejudices. For example, female-dominated vocabulary gets associated with lower-prestige occupations. This is a domain shift problem: the training distribution (biased historical hiring data) differs from the target distribution (equitable outcomes).

Prompt-based Scoring. Evaluators prompt LLMs (like GPT-4) to grade resumes on a 0 to 100 scale. Bias shows up through the model's internal heuristics and RLHF training, which can produce unpredictable behaviors like "DEI" over-correction that favors certain demographics.

Pairwise Ranking. Systems compare two candidates side by side and pick the better fit. This method is especially prone to algorithmic ableism: models penalize resumes with disability related keywords even when those candidates are objectively stronger.

3 Auditing and Detecting Bias

The first step in fairness research is auditing. This section connects the empirical findings to formal group fairness criteria.

3.1 Embedding Retrieval and Intersectional Bias

Wilson and Caliskan (2024) audited large text embedding models with over 3 million cosine similarity comparisons across occupations. Embedding models favored resumes with White-associated names in 85.1% of cases. This is a clear violation of *statistical parity*: $P(\hat{Y} = 1 \mid A = \text{White}) \neq P(\hat{Y} = 1 \mid A = \text{Black})$, where $\hat{Y}$ is the selection outcome and A is the sensitive attribute. Black male candidates were the most penalized group, disadvantaged across nearly all contexts. Document length and name frequency also influenced outcomes, meaning bias flows through multiple channels at once.

3.2 Prompt-based Scoring across Frontier Models

An et al. (2025) took a different approach, auditing five prompt-based LLMs using 361,000 resumes. Their regression model isolated each demographic marker's effect:

$$Score_{i,j,k} = \alpha + \beta \cdot Female + \gamma \cdot BlackMale + \delta \cdot BlackFemale + Position + State + \epsilon \tag{2}$$

where i indexes the resume, j the job position, and k the U.S. state. The intercept α is the baseline score for a White male candidate; β, γ, and δ measure score differences due to gender and race. While embedding models penalize females, prompt-based LLMs slightly preferred female candidates. This tension illustrates the *incompatibility theorem* (Chouldechova, 2017): when base rates differ across groups, no classifier can simultaneously satisfy statistical parity, equalized odds, and predictive rate parity. So a model that favors females for statistical parity may violate equalized odds. Across both mechanisms, one finding was consistent: Black male candidates received lower scores regardless of the architecture.

3.3 Proxy Features

A common defense is removing demographic markers before screening. Bhatnagar et al. (2025) showed this does not work. Bias persists through *proxy features*: career gaps hit women harder because of maternity leave, and word choices ("executed" vs. "helped") encode gender.

Li, Li, and Lu (2023) extended this concept to national origin bias. They mathematically demonstrated how word embeddings exploit vocabulary linked to cultural backgrounds by defining a p-ratio to measure a term's bias:

$$p(t) = \frac{P(t \mid d_{\text{min}})}{P(t \mid d_{\text{max}})} \tag{3}$$

A neutral word like "management" maintains a p-ratio near 1.0, while a culturally revealing word like "Shanghai" drops to 0.02. Without targeted mitigation, models use these culturally specific terms as proxies to penalize certain national origins, resulting in a baseline fairness score of merely 0.309.

3.4 Disability Bias in Pairwise Ranking

Disability bias gets the least attention in AI fairness. Glazko et al. (2024) audited GPT-4's pairwise ranking by adding four disability related achievements (e.g., a disability leadership award) to a standard Control CV, creating an "Enhanced CV" (ECV). The ECV was objectively stronger, yet GPT-4 ranked it first in only 15 out of 60 trials. The model was penalizing candidates for mentioning disability related accommodations or leadership.

3.5 Synthetic Data Validation

Haughey et al. (2025) filled a methodological gap in bias auditing by building a framework for generating synthetic resume datasets with controlled demographic attributes. Their case study showed that synthetic data reproduces the bias patterns found in real-world audits, while letting researchers vary individual features (name, education, experience) to isolate causal effects. This matters because large-scale resume datasets with demographic labels are hard to obtain, which limits reproducibility.

4 Mitigation Strategies: A Layered Defense Pipeline

Because no single method eliminates bias entirely, the mitigation literature is synthesized into a proposed four-layer pipeline, expanding upon Albaroudi et al.'s (2024) recommendation for hybrid pipelines. Figure1 illustrates this architecture.

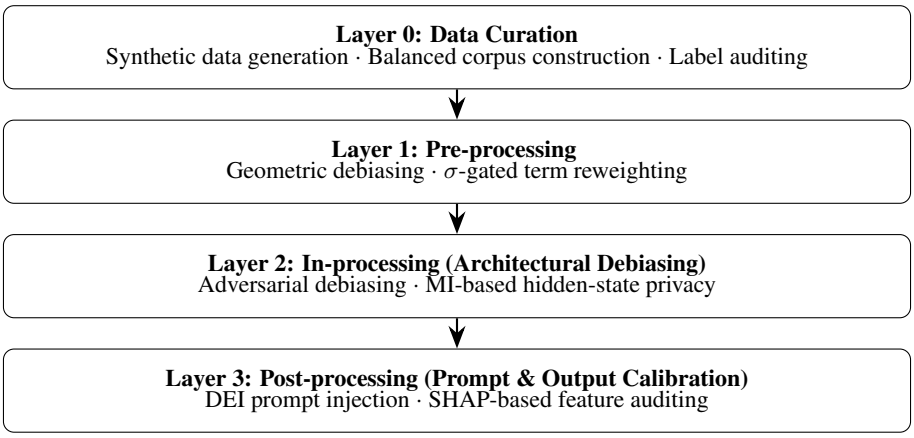


Figure 1: Proposed Layered Pipeline.

4.1 Layer 0: Data Curation

Fairness starts with the training data. Haughey et al. (2025) showed that synthetic resume datasets with controlled demographic distributions can work both as auditing benchmarks and as debiasing training sets. Balancing representation across gender, race, and other protected attributes in the training corpus reduces downstream bias before any model training starts. Mishra (2025) also argued for feature-level auditing with SHAP to find and remove biased proxy features from the dataset before training.

4.2 Layer 1: Pre-processing

Pre-processing interventions clean the data representations before they reach the scoring model.

- **Vector-space debiasing:** Bolukbasi et al. (2016) introduced geometric projection to remove gender directions from word embeddings, ensuring that equations like $\vec{doctor} - \vec{man} + \vec{woman}$ resolve to $\vec{physician}$ rather than biased stereotypes like $\vec{nurse}$.
- **σ-gated term reweighting:** Li et al. (2023) mitigate national origin proxy bias by applying a sigmoid penalty to terms with low p-ratios:

$$\bar{R}(d) = \sum_i f_i \cdot W(t_i) \cdot \sigma(\lambda \cdot (p(t_i) - \tau)) \tag{4}$$

  This effectively zeroes out biased terms (e.g., "Shanghai") while retaining task-relevant terms (e.g., "Python"), successfully increasing overall fairness from 0.309 to 0.782.

4.3 Layer 2: In-processing (Architectural Debiasing)

In-processing methods modify the model's learning objective to mathematically guarantee fairness during training. This layer draws heavily on *adversarial robustness* and *algorithmic fairness*.

Adversarial Debiasing (Zhang et al., 2018): Zhang et al. conceptualized fairness as a two-player game comprising a predictor and an adversary. This formulation directly mirrors the *adversarial training* paradigm, where a robust loss $R_{\text{rob}}(f) = \mathbb{E}[\max_{\|\delta\| \leq \epsilon} \ell(f(x+\delta), y)]$ is minimized via alternating inner maximization and outer minimization. In the fairness setting, the predictor aims to accurately score the resume (generating task loss $\mathcal{L}_{\text{pred}}$), while the adversary tries to deduce the applicant's protected attribute Z from the prediction (generating adversarial loss $\mathcal{L}_{\text{adv}}$). The joint objective minimizes:

$$\min_{\theta_P} \left[\mathcal{L}_{\text{pred}}(\hat{Y}, Y; \theta_P) \; - \; \alpha \cdot \mathcal{L}_{\text{adv}}(\hat{Z}, Z; \theta_A, \theta_P) \right] \tag{5}$$

where θ_P and θ_A denote the predictor and adversary parameters respectively, and α controls the fairness–accuracy trade-off. The gradient update leverages Danskin's theorem to handle the inner maximization, and a gradient projection technique removes any weight updates that would assist the adversary in guessing the demographic, explicitly enforcing constraints like Demographic Parity or Equalized Odds.

Privacy-Enhancing Debiasing (Peña et al., 2025): Peña et al. found that debiasing the final output is not enough if the LLM's internal hidden state (h) still encodes bias. They introduced a mutual information (MI) penalty to the objective function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{task}} + \lambda \cdot I(h; G) \tag{6}$$

where $\mathcal{L}_{\text{task}}$ represents the primary resume scoring loss, $I(h; G)$ quantifies the mutual information between the hidden representation h and the protected attribute G, and λ is a hyperparameter balancing task accuracy against privacy. By minimizing this penalty, the model treats the protected attribute G as private data that the hidden state should not contain. The idea parallels *differential privacy*: the Laplace mechanism adds noise to prevent individual-level inference from query outputs; similarly, the MI penalty strips extractable demographic information from internal representations. After applying this penalty, an adversary's AUC for predicting gender from the hidden state dropped from 0.85 to ≤ 0.55.

4.4 Layer 3: Post-processing (Prompt & Output Calibration)

For closed source API models where weight manipulation is impossible, post-processing techniques are required.

DEI Prompt Injection: Glazko et al. (2024) utilized "Diversity, Equity, and Inclusion (DEI) Prompt Injection," explicitly instructing GPT-4 to value diverse backgrounds and ignore disability related penalizations. This zero-compute method immediately improved the Enhanced CV's top-rank rate from 15/60 to 37/60.

Explainable Feature Auditing: Mishra (2025) proposed using SHAP (SHapley Additive exPlanations) values to post hoc audit model decisions for proxy feature reliance. SHAP is grounded in the *Shapley value* from cooperative game theory: for k input features, the Shapley value $\phi_i(v)$ provides a provably fair attribution of the model's output to each feature i, satisfying efficiency, symmetry, and linearity axioms. By verifying that no protected proxy feature (e.g., career gaps, ZIP codes) ranks among the top contributors to the final score, organizations can flag and reevaluate suspicious predictions before they affect candidates.

Prompt-based methods remain fragile and lack mathematical guarantees compared to in-processing techniques, which is why the full layered pipeline matters.

5 Comparative Analysis and Discussion

5.1 Cross-Method Comparison

Table 1 summarizes the key characteristics of the mitigation strategies examined in this review.

Table 1: Comparison of bias mitigation strategies across four evaluation dimensions.

Strategy	**Math. Guarantee**	**Applicability**	**Compute Cost**	**Limitation**
Data Curation	None	Universal	Low	Cannot fix model-level bias
Vector Debiasing	Geometric	Embedding only	Low	Incomplete removal
σ-gated Reweight	Statistical	Embedding only	Low	Requires p-ratio estimation
Adversarial Debiasing	Min-max	Open-weight	High	Accuracy–fairness trade-off
MI Penalty	Information-theoretic	Open-weight LLM	High	Requires hidden-state access
DEI Prompt	None	Closed-source API	Zero	Fragile, no guarantees
SHAP Audit	Post-hoc	Universal	Medium	Reactive, not preventive

5.2 Acceptance Criteria for Algorithmic Hiring

These 11 papers point to specific acceptance criteria that organizations should require before deploying LLM-based resume screening:

1. **Statistical Parity via Disparate Impact (DI):** The system must satisfy the EEOC's four-fifths rule: $\frac{P(\hat{Y}=1|A=\text{minority})}{P(\hat{Y}=1|A=\text{majority})} \geq 0.8$, directly corresponding to the Disparate Impact Ratio.
2. **Internal Representation Privacy:** Following Peña et al., any adversarial classifier attempting to predict protected attributes from the LLM's hidden states must perform with an AUC ≤ 0.55.
3. **Counterfactual Robustness:** The model must yield rank parity $\geq 1/k$ when processing counterfactually perturbed resumes (where only the demographic proxy is altered). This connects to the *individual fairness* criterion: $d(\hat{Y}_i, \hat{Y}_j) \leq L \cdot d(x_i, x_j)$, ensuring similar individuals receive similar outcomes.
4. **Explainable Feature Attribution:** SHAP values must demonstrate that no protected proxy feature ranks within the top-5 contributors to the final predicted score.

5.3 Limitations and Open Challenges

- **Intersectionality remains understudied.** Most auditing studies examine single demographic axes (e.g., gender *or* race). Only Wilson and Caliskan and An et al. systematically investigate intersectional effects (e.g., Black women versus White men), revealing that intersectional penalties are often nonadditive. The disadvantage of being both Black and male exceeds the sum of individual penalties.
- **The accuracy and fairness trade-off is unavoidable.** Theoretical formulations prove that for any classifier satisfying statistical parity, the combined error satisfies $\epsilon_{A=0}(h) + \epsilon_{A=1}(h) \geq \Delta_{\text{BR}}$, where Δ_{BR} is the base rate difference between groups. This theoretical lower bound means that enforcing fairness necessarily incurs some accuracy loss, a tension that no surveyed mitigation strategy fully resolves.
- **Temporal drift as distribution shift.** No surveyed paper addresses how model bias evolves across fine-tuning updates or RLHF iterations. This is a *domain adaptation* problem: the source domain diverges from the target domain, and the $\mathcal{H}$-divergence between these domains may grow without continuous monitoring.

6 Conclusion and Future Work

LLMs in resume screening offer scalability but risk baking in historical biases. As the papers reviewed here show, bias is structural, and surface-level fixes like name removal do not work against proxy features. A fair resume screening system requires a combination of techniques. Fairness alone fails without robustness to adversarial proxy features; privacy-preserving representations are useless without interpretable auditing; and all interventions break down without generalization guarantees across distribution shifts. A layered defense pipeline that combines data curation, vector-space, pre-processing, adversarial in-processing, and prompt injection can reduce these risks substantially.

Several research directions stand out. First, *standardized bias benchmarks* for resume screening would allow reproducible cross-model comparisons. Second, *continuous monitoring frameworks* are needed to detect bias drift across model updates, formalized as a domain adaptation problem where the $\mathcal{H}$-divergence between audit-time and deployment-time distributions must stay bounded. Third, auditing methods should extend to *additional protected categories* like age, socioeconomic background, and neurodivergence. Fourth, *bias-auditing APIs* could let external parties verify closed source models without exposing proprietary weights, balancing the privacy-utility trade-off from differential privacy.

Building on this review, a key direction for future work is the practical implementation of the proposed layered defense pipeline. Developing a functional prototype that integrates data curation, adversarial debiasing, and prompt calibration will allow for empirical testing of how these layers interact to reduce bias in real-world scenarios. Ultimately, algorithmic fairness requires mathematical constraints, continuous auditing, regulatory alignment, and an honest recognition that technology alone will not fix deeply rooted societal inequitie

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